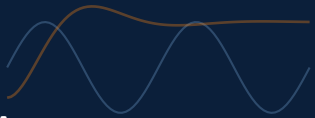
DAY 5 / 5

Mechatronics: Electrical Machines, Control, Servo Systems and Sensing



Resonance, Sensors, Measurement and System Integration

3.7 Resonance & filters ▪ 4.1 Sensors ▪ 4.2 Encoders & resolvers ▪ 4.3 Sampling ▪ 4.4 Signal processing ▪ 4.5 Communication

Mechatronics Training Programme

Theoretical Course Series ▪ Day 5 of 5

Agenda and learning objectives

Code	Session	Time
3.7	Resonance, vibration and filter use	60 min
4.1	Sensors and measurement principles	30 min
4.2	Encoder and resolver	45 min
4.3	Sampling and digitisation	30 min
4.4	Signal processing and filters	90 min
4.5	Communication and system integration	45 min
	Course wrap-up (inside 4.5 block)	

5 h net teaching time; 10-min breaks roughly every hour are extra.

After today you can . . .

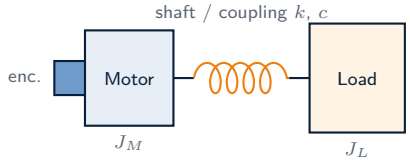
- model a compliant load and predict f_{AR}, f_R
- size a notch / low-pass torque filter
- choose and interpret sensors and encoders
- quantify velocity quantisation and ADC noise
- select FIR / IIR filters and implement them
- budget latency and jitter on a fieldbus
- connect all five days into one servo picture

Session 1

Resonance, vibration and filter use (3.7)

Why mechanical resonance limits servo bandwidth

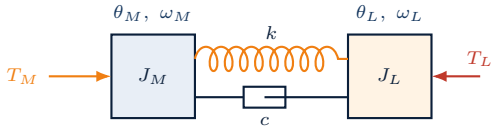
- Days 3–4 assumed a **rigid** load: $G(s) = 1/(J_{\text{tot}}\, s)$.
- Real drive trains contain **compliance**: shafts, couplings, belts, ball screws, gearboxes, frames.
- Compliance + inertia = **lightly damped mode** ($\zeta \approx 0.01$–0.1).
- The velocity loop sees the mode through the motor encoder: a **gain peak** at f_{R} and a dip at f_{AR}.
- Raising the gain to stiffen the axis pushes the peak above 0 dB $\Rightarrow$ **audible ringing / instability**.



Typical f_{R}: belts 20 Hz to 100 Hz, ball screws 100 Hz to 500 Hz, couplings / shafts 0.3 kHz to 3 kHz

Feedback on the motor side is **collocated**: the mode can be destabilised by gain, but its phase stays within $\pm 90°$ of the rigid case — notch filtering is effective.

Two-mass model: equations of motion



$$J_M\,\dot{\omega}_M = T_M - T_S$$
$$J_L\,\dot{\omega}_L = T_S - T_L$$
$$T_S = k(\theta_M - \theta_L) + c(\omega_M - \omega_L)$$

k [N m/rad] torsional stiffness, c [N m s/rad] damping, T_S shaft torque. Reflect through a gear i: $J_L' = J_L/i^2$, $k' = k/i^2$.

Laplace transform with $T_L = 0$:

$$G_M(s) = \frac{\omega_M}{T_M} = \frac{J_L s^2 + cs + k}{s\,\Delta(s)}$$
$$G_L(s) = \frac{\omega_L}{T_M} = \frac{cs + k}{s\,\Delta(s)}$$
$$\Delta(s) = J_M J_L s^2 + c(J_M{+}J_L)s + k(J_M{+}J_L)$$

- G_M: **zero pair** (anti-resonance) + **pole pair** (resonance).
- G_L: only the pole pair, phase drops to $-270°$ (**non-collocated**).
- Low f: $G_M \to 1/((J_M{+}J_L)s)$; high f: $G_M \to 1/(J_M s)$.

Resonance and anti-resonance frequencies

$$\omega_{\mathrm{AR}} = \sqrt{\frac{k}{J_L}}, \qquad \omega_{\mathrm{R}} = \sqrt{\frac{k\,(J_M + J_L)}{J_M\,J_L}}$$

$$\frac{\omega_{\mathrm{R}}}{\omega_{\mathrm{AR}}} = \sqrt{1+R}, \qquad R = \frac{J_L}{J_M}$$

- AR: the load oscillates against a *motionless* motor (motor appears clamped).
- R: motor and load oscillate in anti-phase around the common centre of inertia.
- Gain step between the asymptotes: $20\log_{10}(1+R)$ dB.
- Damping ratios: $\zeta_{\mathrm{AR}} = \dfrac{c}{2\sqrt{kJ_L}}$, $\zeta_{\mathrm{R}} = \zeta_{\mathrm{AR}}\sqrt{1+R}$.

Worked example

$J_M = 1.2\times10^{-4}\,\mathrm{kg\,m^2}$, $J_L = 6.0\times10^{-4}\,\mathrm{kg\,m^2}$ ($R = 5$), coupling $k = 600\,\mathrm{N\,m/rad}$.

$\omega_{\mathrm{AR}} = \sqrt{600/6\times10^{-4}} = 1000\,\mathrm{rad/s} \Rightarrow f_{\mathrm{AR}} = 159\,\mathrm{Hz}$

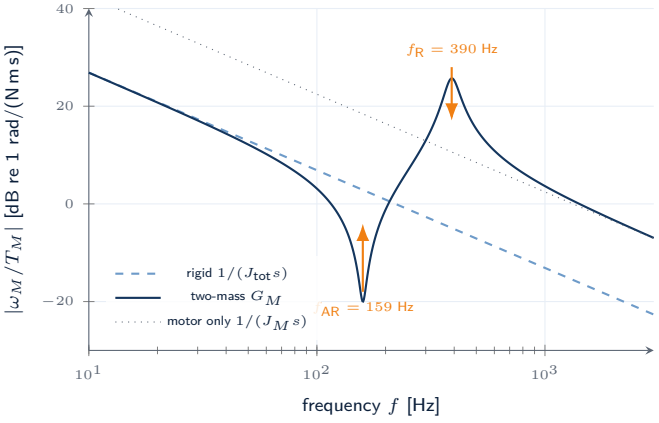
$\omega_{\mathrm{R}} = 1000\sqrt{6} = 2449\,\mathrm{rad/s} \Rightarrow f_{\mathrm{R}} = 390\,\mathrm{Hz}$

$\Delta G = 20\log_{10}6 = 15.6\,\mathrm{dB}$

With $c = 0.036\,\mathrm{N\,m\,s/rad}$: $\zeta_{\mathrm{AR}} = 0.03$, $\zeta_{\mathrm{R}} = 0.073$.

A 15.6 dB higher plant gain above f_{R} means a loop designed for the total inertia has its crossover pushed up by $\times6$ if the gain is not reduced there.

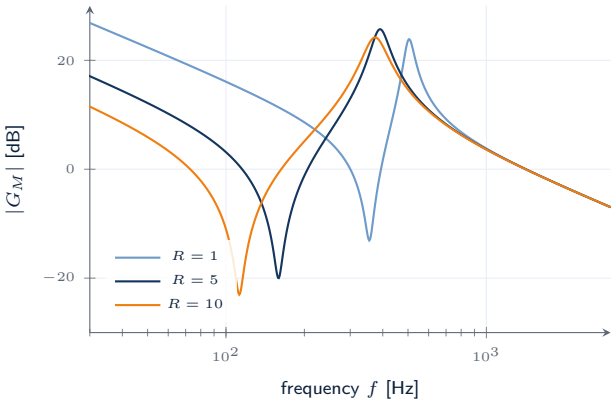
Frequency response of the two-mass system



- Below f_{AR}: the whole drive train moves as one body.
- Between f_{AR} and f_R: the load "decouples".
- Above f_R: the motor alone is accelerated, gain 15.6 dB higher.
- The **peak height** is set by ζ_R — typically only a few % in metal drive trains.

Computed from $G_M(s)$ with the values of the previous slide.

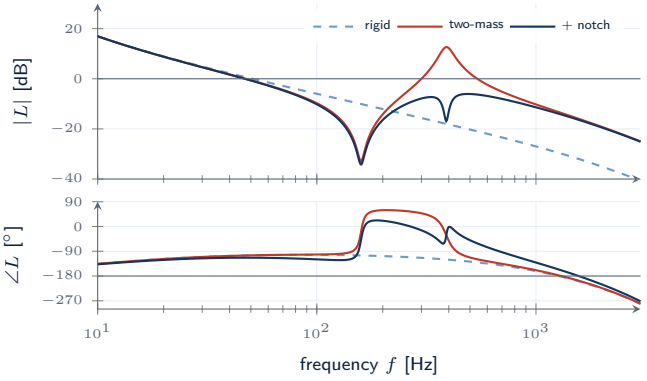
Effect of the inertia ratio $R = J_L/J_M$



J_M, k fixed; $\zeta_{\mathsf{AR}} = 0.03$.

- Larger $R \Rightarrow f_{\mathsf{AR}}$ moves down, $f_{\mathsf{R}}/f_{\mathsf{AR}} = \sqrt{1+R}$ widens.
- The rigid-body gain falls with $1/(1+R)$, the gain above f_{R} does not.
- Rule of thumb: $R \lesssim 3$–5 for stiff high-bandwidth axes, $R \lesssim 10$ for general servo work; beyond that tuning becomes difficult.
- Gear ratio i reduces the reflected ratio: $R' = J_L/(i^2 J_M)$; **inertia matching** $i_{\mathsf{opt}} = \sqrt{J_L/J_M}$ maximises load acceleration.

Effect on the velocity-loop open-loop FRF



PI velocity loop, $f_c = 50$ Hz, current loop 2 kHz, delay 125 µs; plant: $R = 5$ example.

- Without filter the resonance lifts $|L|$ to **+12.6 dB** at 390 Hz: extra 0-dB crossings ⇒ the axis rings / goes unstable.
- Notch ($\zeta_z = 0.02$, $\zeta_p = 0.6$) at f_R: peak −17 dB, GM ≈ 9 dB.
- Price: phase margin $75° \to 66.5°$ at 47 Hz (notch lag ≈ $8.6°$).

The biquad notch filter

$$N(s) = \frac{s^2 + 2\zeta_z\,\omega_n s + \omega_n^2}{s^2 + 2\zeta_p\,\omega_n s + \omega_n^2}, \qquad \zeta_z < \zeta_p$$

- **Centre frequency** ω_n: place at the measured ω_R.
- **Depth** at ω_n: $|N(j\omega_n)| = \zeta_z/\zeta_p$, i.e. $20\log_{10}(\zeta_z/\zeta_p)$ dB.
- **Width**: for a deep notch the -3 dB band is $\Delta\omega \approx 2\zeta_p\omega_n$ (quality $Q = 1/(2\zeta_p)$).
- Unity gain at DC and at high frequency; phase lag *below* ω_n, lead above.
- Lag at crossover ($\omega_c \ll \omega_n$): $\Delta\varphi \approx -\dfrac{2(\zeta_p - \zeta_z)\,\omega_c/\omega_n}{1 - (\omega_c/\omega_n)^2}$ rad.

Digital form (Tustin with pre-warping)

$$K = \omega_n/\tan(\omega_n T_s/2),\;\; D = K^2 + 2\zeta_p\omega_n K + \omega_n^2$$

$$b_0 = (K^2 + 2\zeta_z\omega_n K + \omega_n^2)/D$$

$$b_1 = a_1 = 2(\omega_n^2 - K^2)/D$$

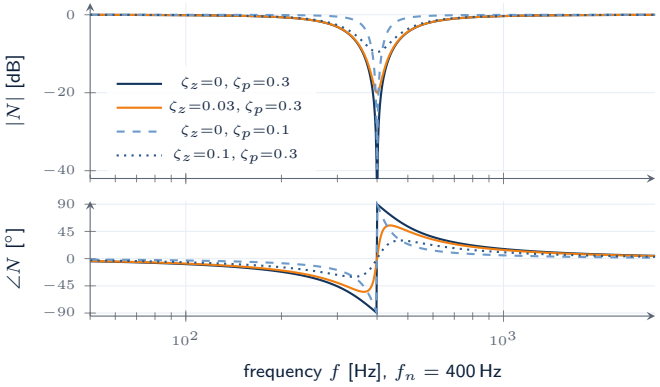
$$b_2 = (K^2 - 2\zeta_z\omega_n K + \omega_n^2)/D$$

$$a_2 = (K^2 - 2\zeta_p\omega_n K + \omega_n^2)/D$$

$$N(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

Pre-warping keeps the notch exactly at ω_n although Tustin compresses the frequency axis.

Notch filter: width and depth in the Bode plot



- $\zeta_z = 0$: infinitely deep notch, phase jumps by 180° at f_n.
- $\zeta_z/\zeta_p = 0.1 \Rightarrow -20\,\text{dB}$, $1/3 \Rightarrow -9.5\,\text{dB}$.
- Narrow notch ($\zeta_p = 0.1$) costs little phase at low f but must hit f_R precisely.
- Wide notch is **robust** to f_R drift (load, position, temperature) but costs more phase at ω_c.

Selecting notch parameters: a practical procedure

1. Measure the open-loop / plant FRF (sine sweep or PRBS, Day 2 identification) at low gain.
2. Read f_R from the peak; check spread over the axis stroke and payload.
3. Set $f_n = f_R$; choose depth $\geq$ peak height above the desired gain line plus 6 dB to 10 dB.
4. Choose width: $\zeta_p \approx 0.3$–0.7 if f_R varies by $\pm 10\,\%$ to 20 %, else narrower.
5. Re-measure; check PM at ω_c (notch lag) and GM at f_R.
6. Raise the velocity gain until the next limit (next mode, delay, noise).

Phase cost of the example notch

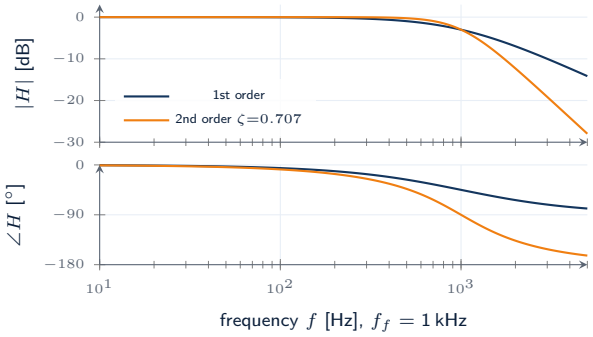
$f_n = 390$ Hz, $\zeta_p = 0.6$, $\zeta_z = 0.02$, $f_c = 50$ Hz, $x = f_c/f_n = 0.128$:

$$\Delta\varphi = -\arctan\frac{2(0.6)(0.128)}{1-0.128^2} + \arctan\frac{2(0.02)(0.128)}{1-0.128^2}$$

$$\Delta\varphi \approx -8.9^\circ + 0.3^\circ = -8.6^\circ$$

Rule: keep $f_n \gtrsim 3$–$5\,f_c$. A notch closer to crossover eats the phase margin and should be replaced by mechanical stiffening or a lower f_c.

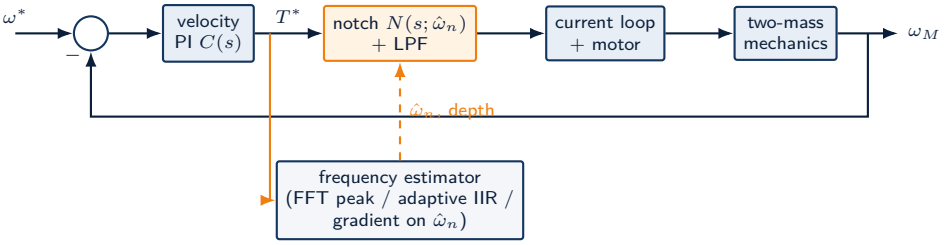
Low-pass torque (current-reference) filters



$$H_1 = \frac{\omega_f}{s + \omega_f}, \quad H_2 = \frac{\omega_f^2}{s^2 + 2\zeta\omega_f s + \omega_f^2}$$

- Attenuate **broad-band** high-frequency content: many modes above $\sim 5 f_c$, encoder quantisation noise, current ripple.
- Phase cost at $f_c = 100$ Hz, $f_f = 1$ kHz: H_1: $-5.7°$; H_2: $-8.1°$.
- Attenuation at 390 Hz only -0.6 dB (H_1) $\Rightarrow$ a LPF *cannot* fix a resonance close to f_c; a notch can.
- Typical: $f_f \approx (5\text{–}10)\, f_c$; 2nd order when noise matters.

Adaptive notch: tracking a moving resonance



- Monitor the torque reference (or velocity error) for a dominant narrow-band component above f_c.
- Estimate its frequency online and move ω_n; freeze adaptation when the signal is small.
- Needed when f_{R} changes with payload, position (long screw) or belt tension.
- Risks: adaptation on a *forced* disturbance, slow convergence, interaction of two notches — always bound the range.

Summary: resonance and filter use

- Compliance turns $1/(Js)$ into $G_M(s)$ with an anti-resonance $\sqrt{k/J_L}$ and a resonance $\sqrt{1+R}$ times higher.
- High inertia ratio $\Rightarrow$ large gain step and wide separation $\Rightarrow$ harder tuning.
- The **notch** removes a narrow peak at small phase cost if $f_n \gtrsim 3$–$5 f_c$.
- The **low-pass** handles broad high-frequency content; it costs phase everywhere below f_f.
- Adaptive notches track drifting modes; bound and supervise them.

Design order

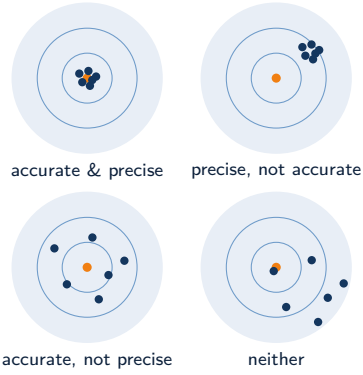
1. Stiffen mechanics, reduce R
2. Measure FRF
3. Notch dominant modes, LPF the rest
4. Re-tune $C(s)$, verify PM/GM
5. Re-check over the operating envelope

Further reading: [Ellis, 2012], ch. on resonance; [Franklin, Powell & Emami-Naeini, 2019].

Session 2

Sensors and measurement principles (4.1)

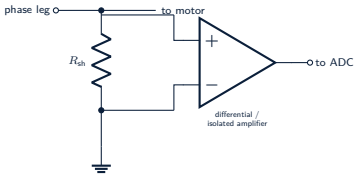
Metrology vocabulary



Term	Meaning
Accuracy	closeness to the true value (systematic + random error)
Precision	spread of repeated readings (σ)
Repeatability	precision under identical short-term conditions
Resolution	smallest detectable change (e.g. 1 LSB, 1 count)
Range	span between min. and max. measurable value
Linearity	max. deviation from best-fit line, % of full scale

> High resolution $\neq$ high accuracy: a 23-bit encoder (0.15 arcsec/count) may have 20 arcsec absolute error.

Current measurement: shunt, Hall and fluxgate



$$u_{sh} = R_{sh}\, i, \quad P = R_{sh}\, I_{rms}^2$$

Example

$\hat{I} = 20\,\text{A}$, $R_{sh} = 2\,\text{m}\Omega$: $\hat{u}_{sh} = 40\,\text{mV}$; at $I_{rms} = 10\,\text{A}$, $P = 0.2\,\text{W}$. Gain 20 $\Rightarrow$ 0.8 V at the ADC.



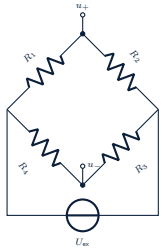


SMD shunts: wdwd, CC BY-SA 4.0; Hall IC: SparkFun, CC BY 2.0; via Wikimedia Commons

	Shunt	Hall OL / CL	Fluxgate
Isolation	no (needs isol. amp / $\Sigma\Delta$)	yes	yes
Accuracy	0.1–0.5 %	1–3 % / 0.2–0.5 %	$<$0.1 %
Offset drift	very low	medium	very low
Bandwidth	DC–MHz	DC–50 kHz to 300 kHz	DC–10 kHz to 200 kHz
Losses	I^2R	small	small
Cost / size	lowest	medium	highest

- Hall: $u_H \propto B \propto i$ in a slotted core; closed loop nulls the flux with a compensation winding.
- Fluxgate: a periodically saturated core detects zero flux $\Rightarrow$ very low offset.
- Offset in i_a, i_b gives a torque ripple at f_{el}; gain mismatch at $2f_{el}$.

Force: strain gauges and load cells





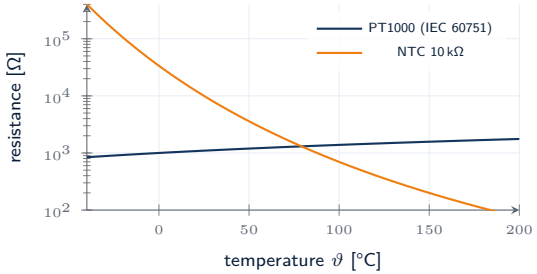


Foil gauge: Pleriche, CC BY-SA 4.0; load cell: Daraceleste, CC BY-SA 4.0; via Wikimedia Commons

$$\frac{\Delta R}{R} = k_{\mathsf{GF}}\,\varepsilon, \qquad \frac{u_{\mathsf{out}}}{U_{\mathsf{ex}}} \approx \frac{k_{\mathsf{GF}}\,\varepsilon}{4}\,n_{\mathsf{active}}$$

- Metal foil gauge: gauge factor $k_{\mathsf{GF}} \approx 2$; strains are μ-level.
- **Full bridge** (4 active arms, $\pm\varepsilon$): 4× sensitivity, temperature compensation, rejects bending.
- Example: $\varepsilon = 1000\,\mu\epsilon$, $U_{\mathsf{ex}} = 5\,\mathrm{V}$, full bridge: $u_{\mathsf{out}} = 2 \cdot 10^{-3} \cdot 5\,\mathrm{V} = 10\,\mathrm{mV}$.
- Load cells are specified by rated output, e.g. 2 mV/V at nominal force ⇒ need a low-noise instrumentation amplifier or 24-bit $\Sigma\Delta$ ADC.
- Errors: creep, hysteresis, temperature of zero and span, non-linearity (∼0.02–0.1 % FS).

Temperature: NTC, PT100/PT1000, thermocouple, KTY







Left: wire-wound Pt100, Harke, CC BY-SA 3.0; right: type-K thermocouple, Harke, public domain; both via Wikimedia Commons

PT100/PT1000 ($\vartheta \geq 0\,°\mathrm{C}$, Callendar–Van Dusen):

$$R = R_0(1 + A\vartheta + B\vartheta^2)$$

$$A{=}3.9083{\cdot}10^{-3}\,\mathrm{K}^{-1},\ B{=}{-}5.775{\cdot}10^{-7}\,\mathrm{K}^{-2}$$

NTC: $R = R_{25}\exp\left[B\,(1/T - 1/T_{25})\right]$, T in K.

- **PT**: linear, accurate (class A: $\pm(0.15 + 0.002|\vartheta|)$ K), 3/4-wire.
- **NTC**: very sensitive, non-linear; IGBT modules, heatsinks.
- **Thermocouple**: Seebeck voltage, type K $\approx 41\,\mu\mathrm{V/K}$, needs cold-junction compensation.
- **KTY** (silicon PTC, e.g. KTY84: 1 kΩ at 100 °C): motor winding, used for thermal models; PTC triplets for trip only.

Summary: sensors and measurement principles

- Specify sensors with the full vocabulary: accuracy, precision, resolution, range, repeatability, linearity.
- Currents: shunt (cheap, accurate, needs isolation), Hall (isolated, drift), fluxgate (precision).
- Force: strain-gauge bridges deliver mV/V — amplifier noise and drift dominate.
- Temperature: PT for accuracy, NTC for sensitivity, thermocouple for high ϑ, KTY in motors.

Checklist for any sensor

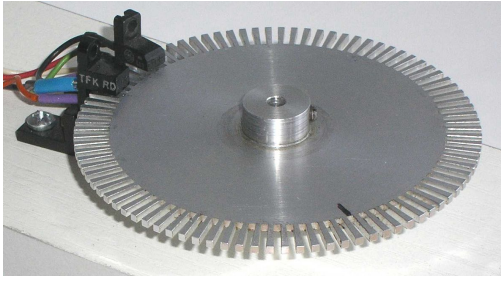
- What error matters: offset, gain, noise or drift?
- Bandwidth and latency vs. loop f_c
- Isolation and EMC environment
- Calibration and diagnostics

[Fraden, 2016]; IEC 60751; manufacturer notes (TI, ADI, LEM-type sensor datasheets).

Session 3

Encoder and resolver (4.2)

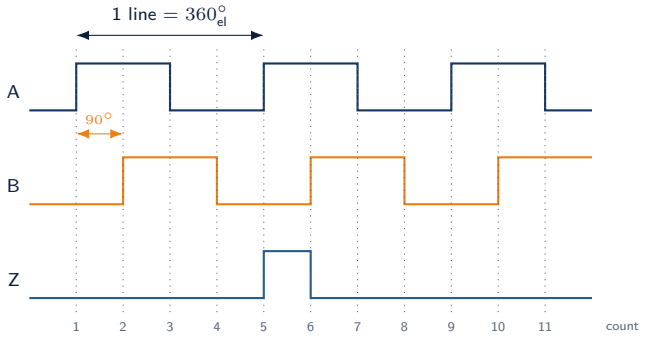
Optical incremental encoders



Rrudzik, CC BY-SA 4.0, via Wikimedia Commons

- LED → code disk with N lines (PPR) → phase-shifted gratings → photodiodes.
- Outputs: **A**, **B** (90° shifted) and **Z** (index, once per rev.), usually RS-422 differential.
- Edge counting gives position relative to power-up ⇒ **homing** needed.
- Max. output frequency $f_A = N\,n/60$; e.g. $N = 2500$, $n = 6000$ rpm ⇒ 250 kHz (1 MHz edge rate).
- Also magnetic (Hall / MR) and inductive variants: robust, lower resolution.

Quadrature signals and $\times 4$ decoding



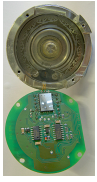
prev. AB	new AB	count
00	10	$+1$
10	11	$+1$
11	01	$+1$
01	00	$+1$
reverse order		-1
both change		error

- Every edge of A and B is counted: $4N$ counts/rev.
- Direction from which channel leads (A leads $\Rightarrow$ forward).
- Both bits changing at once = lost count $\Rightarrow$ error flag.
- Digital glitch filter limits max. edge rate.

Absolute encoders, Gray code and serial protocols



3-bit Gray-code disk

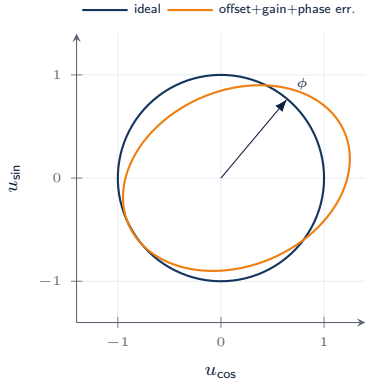


13-track Gray-code encoder, Mike1024, public domain, via Wikimedia Commons

- **Single-turn**: unique code per angle, e.g. 23 bit $= 8\,388\,608$ positions/rev. **Multi-turn**: + 12–16 bit revolution counter (gear stages, battery or Wiegand energy harvesting).
- **Gray code**: only one bit changes per step $\Rightarrow$ no false codes at transitions; $g = b \oplus (b \gg 1)$.

Protocol	Physical	Features
SSI	RS-422, clock + data	unidirectional, simple, no CRC by default, $\sim$1 MHz
BiSS-C	RS-422	open, bidirectional registers, CRC, line-delay comp., up to 10 MHz
EnDat 2.2	RS-485	bidirectional, CRC, diagnostics, electronic ID label, up to 16 MHz
One-cable (DSL)	RS-485 in motor cable	digital link via power cable, saves connector

Sin/cos encoders and interpolation



- Analogue tracks: $u_{\sin} = \hat{U}\sin(N\theta)$, $u_{\cos} = \hat{U}\cos(N\theta)$, typ. $1\,V_{pp}$, $N = 512$–2048 periods/rev.
- Coarse position: count periods; fine position inside one period:

$$\theta = \frac{2\pi}{N}\left(m + \frac{\operatorname{atan2}(u_{\sin}, u_{\cos})}{2\pi}\right)$$

- $N = 2048$ with 12-bit interpolation $\Rightarrow 2^{23}$ counts/rev.
- Error harmonics per signal period: **offset → 1st, amplitude mismatch, phase error → 2nd**; corrected by online calibration of the Lissajous ellipse.
- Interpolation accuracy is limited by signal quality, noise and ADC resolution, not by the arithmetic.

Resolver: principle and signals

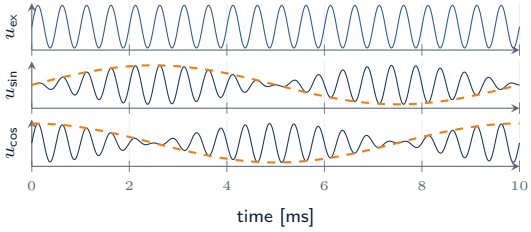
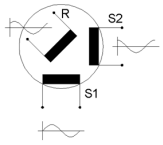


Illustration: carrier 2 kHz (real: 5 kHz to 20 kHz), θ ramping at 100 rev/s.



Hasee, public domain, via Wikimedia Commons

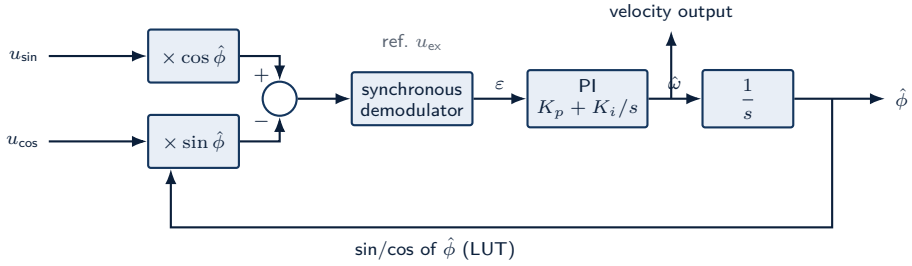
- Rotary transformer: rotor winding R excited by $u_{\text{ex}} = \hat{U} \sin \omega_e t$.
- Two stator windings S1, S2 90° apart:

$$u_{\text{sin}} = k_r \hat{U} \sin(\omega_e t)\ \sin(p_r \theta)$$

$$u_{\text{cos}} = k_r \hat{U} \sin(\omega_e t)\ \cos(p_r \theta)$$

- Angle is in the **envelopes** (dashed); $k_r \approx 0.5$, p_r = speed (pole-pair) count.
- No electronics in the motor: robust to 150 °C+, shock, vibration; typ. accuracy 5 arcmin to 10 arcmin.

Resolver-to-digital conversion: the tracking loop

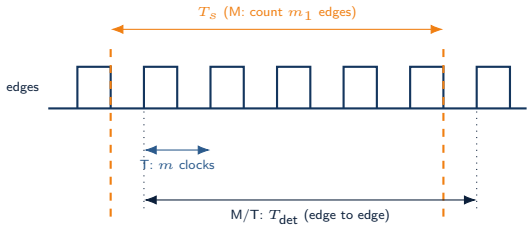


$$\varepsilon \propto \sin p_r\theta \cos\hat{\phi} - \cos p_r\theta \sin\hat{\phi} = \sin(p_r\theta - \hat{\phi})$$

- For small errors $\varepsilon \approx p_r\theta - \hat{\phi}$: a **type-II** PLL.

- Zero steady-state error at constant speed; acceleration error $\propto \alpha/K_a$.
- Loop bandwidth (typ. 1 kHz to 5 kHz) trades noise vs. lag.
- Integrated RDC ICs (e.g. AD2S1210-class) or software RDC in the drive MCU / FPGA.

Velocity from position: M-, T- and M/T-method



	good at	weak at
M	high speed	low speed (few counts)
T	low speed	high speed (few clocks), no edge $\Rightarrow$ no update
M/T	whole range	needs edge time-stamping HW

With $4N$ counts/rev, clock f_{clk}:

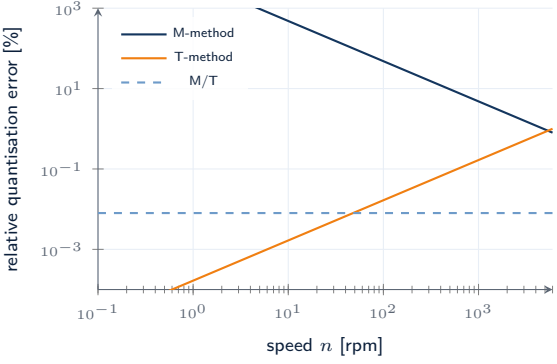
$$\text{M:}\quad \omega = \frac{2\pi\, m_1}{4N\, T_s}$$

$$\text{T:}\quad \omega = \frac{2\pi}{4N} \cdot \frac{f_{\text{clk}}}{m}$$

$$\text{M/T:}\quad \omega = \frac{2\pi\, m_1}{4N} \cdot \frac{f_{\text{clk}}}{m_2}$$

- M/T: count m_1 edges and m_2 clocks over a window that starts and ends on an edge (window $\geq T_s$).
- Modern drives: **edge time-stamping** + absolute high-resolution position.

Low-speed limits and quantisation noise



$N = 2500$ lines, $T_s = 125\,\mu s$, $f_{clk} = 100\,MHz$.

Worked example

M-method resolution:

$$\Delta n = \frac{60}{4N\,T_s} = \frac{60}{10^4 \cdot 125 \cdot 10^{-6}} = 48\ \text{rpm}.$$

At 60 rpm the raw estimate jumps between 48 and 96 rpm!

Differencing uniform position noise ($\sigma_\theta = q/\sqrt{12}$):

$$\sigma_\omega = \frac{\sqrt{2}\,q}{\sqrt{12}\,T_s} = 0.41\,\frac{q}{T_s} \approx 20\ \text{rpm}$$

23-bit absolute encoder: $\Delta n = 0.057$ rpm.

- Remedies: higher resolution, longer window, edge timing, observer or filter (next sessions).

Summary: encoder and resolver

- Incremental A/B/Z: $\times 4$ decoding, homing required, simple and fast.
- Absolute single/multi-turn: Gray code internally, serial protocols (SSI, BiSS-C, EnDat) with CRC.
- Sin/cos: interpolation by atan2; offset/gain/phase errors produce 1st/2nd harmonics.
- Resolver: robust analogue sensor; RDC = type-II tracking loop.
- Velocity estimation is a **quantisation** problem at low speed.

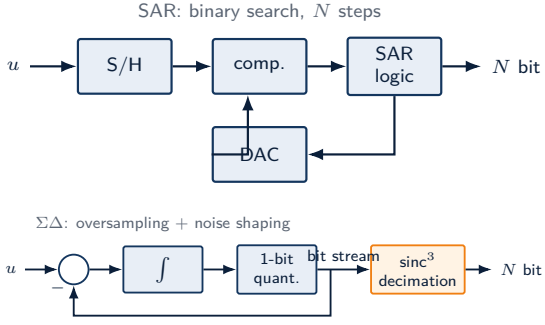
	incr.	**abs.**	**resolver**
Resolution	med.	high	med.
Absolute	no	yes	1 turn
Robustness	med.	med.	high
Diagnostics	low	high	med.
Cost	low	high	med.

[Krishnan, 2010]; ADI and TI resolver / encoder application notes; EnDat, BiSS and SSI interface specifications.

Session 4

Sampling and digitisation (4.3)

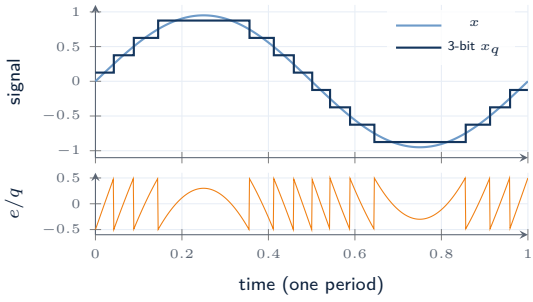
ADC architectures for drives: SAR and sigma-delta



	SAR	Sigma-delta
Resolution	12–16 bit	16–24 bit
Rate	1–5 MS/s	kS/s–MS/s (after decim.)
Latency	1 conversion	filter group delay
Sampling instant	sharp (S/H)	averaged over window
Typical use	phase currents, on-chip MCU ADC	isolated current meas., load cells, temperatures

Isolated $\Sigma\Delta$ modulators next to the shunt + sinc filter in MCU/FPGA are standard for high-performance current measurement.

Quantisation and SNR $= 6.02N + 1.76$ dB



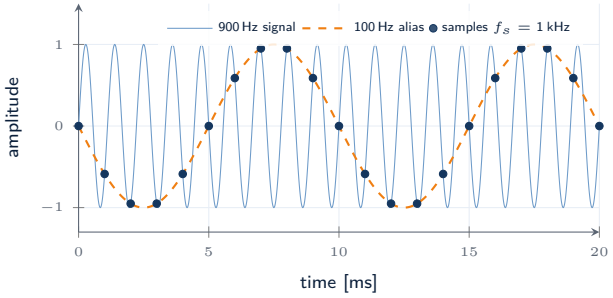
Step $q = 2U_{\text{FS}}/2^N$; error uniform in $[-q/2, q/2]$:

$$P_e = \frac{q^2}{12}, \qquad P_s = \frac{1}{2}\left(\frac{2^N q}{2}\right)^2 = \frac{2^{2N} q^2}{8}$$

$$\text{SNR} = 10\log_{10}\frac{P_s}{P_e} = 10\log_{10}\left(1.5 \cdot 2^{2N}\right) = 6.02N + 1.76\ \text{dB}$$

- 12 bit: 74 dB; 16 bit: 98 dB (full-scale sine).
- Real ADCs: ENOB $=$ (SINAD $- 1.76)/6.02$, often 1–2 bit below N.
- Example: ± 50 A range, 12 bit $\Rightarrow$ LSB $= 24.4$ mA.
- Oversampling by M + averaging: $+10\log_{10} M$ dB if noise is white.

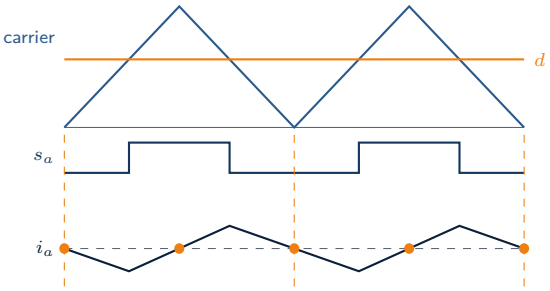
Aliasing and the Nyquist criterion



$$f_{\text{alias}} = |f - k\,f_s|\,, \quad k = \text{round}(f/f_s)$$

- Nyquist: a signal is uniquely represented only if $f_{\text{max}} < f_s/2$.
- Content above $f_s/2$ **folds** into the base band and is indistinguishable from real low-frequency signals.
- Classical drive example: PWM ripple, encoder interpolation errors or a resonance aliasing onto the velocity loop.
- Once sampled, aliasing *cannot* be removed by any digital filter.

Synchronous sampling with PWM, timestamps and latency



- Centre-aligned PWM: sampling at the carrier peak/valley returns the **average** current without ripple or switching noise ⇒ no extra filter.
- Single or double update ($f_s = f_{\text{PWM}}$ or $2f_{\text{PWM}}$).
- Time-stamp every sample (encoder, current, bus) on a common time base.

Delay → phase

$f_s = 16\,\text{kHz}$, $T_d = 1.5T_s = 94\,\mu\text{s}$: phase lag at 1 kHz $360° \cdot f\,T_d = 34°$ — comparable to the whole PM budget of the current loop.

Summary: sampling and digitisation

- SAR for low latency, $\Sigma\Delta$ for resolution and isolation.
- Ideal quantisation: SNR $= 6.02N + 1.76$ dB; check ENOB in practice.
- Anything above $f_s/2$ folds back: filter *before* sampling.
- Sample synchronously with PWM; time-stamp all measurements.
- Every sample of delay is phase lag: $360^\circ f T_d$.

Remember

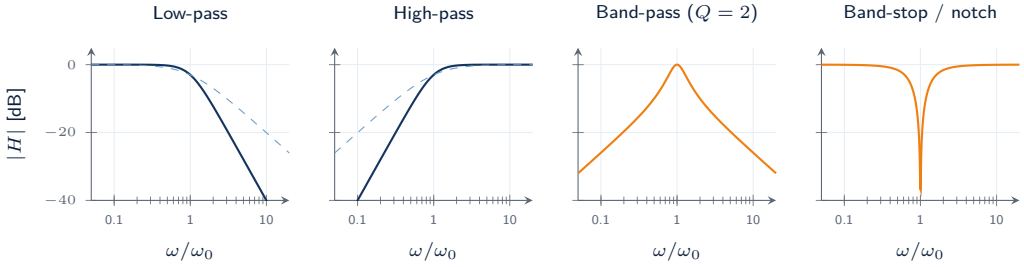
Resolution, noise, bandwidth and delay of the measurement chain set the achievable loop bandwidth — not the controller structure.

[Oppenheim & Schafer, 2010]; [Franklin, Powell & Workman, 1998]; ADI/TI ADC application notes.

Session 5

Signal processing and filters (4.4)

Filter types at a glance



Type	2nd-order prototype	Typical use in a drive
LPF	$\omega_0^2/(s^2 + 2\zeta\omega_0 s + \omega_0^2)$	noise on velocity / torque reference, anti-aliasing
HPF	$s^2/(s^2 + 2\zeta\omega_0 s + \omega_0^2)$	remove offset / drift (AC coupling), wash-out in feed-forward
BPF	$2\zeta\omega_0 s/(s^2 + 2\zeta\omega_0 s + \omega_0^2)$	extract a known frequency (vibration monitoring, resolver carrier)
BSF	$(s^2 + \omega_0^2)/(s^2 + 2\zeta\omega_0 s + \omega_0^2)$	resonance suppression, mains / ripple rejection

First-order low-pass: from $H(s)$ to a difference equation

$$H(s) = \frac{\omega_f}{s + \omega_f}$$

- **Exact (ZOH / impulse-invariant)**:

$$y[k] = y[k-1] + \alpha\,\big(x[k] - y[k-1]\big)$$
$$\alpha = 1 - e^{-\omega_f T_s}$$

- **Tustin**: $s \to \frac{2}{T_s}\frac{z-1}{z+1}$, matches the frequency response better near $f_s/2$.
- First-order HPF: $y_{\text{HP}}[k] = x[k] - y_{\text{LP}}[k]$ (complementary pair).
- Small $\omega_f T_s$: $\alpha \approx \omega_f T_s$ (**exponential smoothing**).

Worked example

$f_f = 500\,\text{Hz}$, $f_s = 16\,\text{kHz}$ ($T_s = 62.5\,\mu\text{s}$):

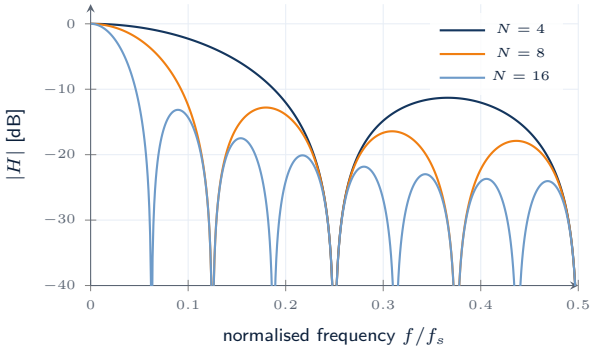
$$\omega_f T_s = 2\pi \cdot 500 \cdot 62.5 \cdot 10^{-6} = 0.196$$

$$\alpha = 1 - e^{-0.196} = 0.178$$

Group delay at low f: $1/\omega_f = 318\,\mu\text{s} \approx 5$ samples.

Every low-pass filter in a loop is a delay of roughly $1/\omega_f$ at low frequency: include it in the loop model.

Moving average: a simple FIR filter



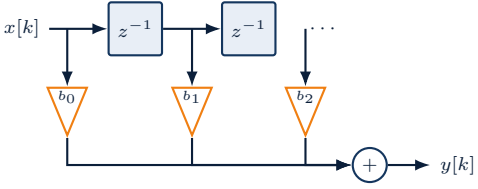
$$y[k] = \frac{1}{N} \sum_{i=0}^{N-1} x[k-i]$$

$$|H(e^{j\Omega})| = \left| \frac{\sin(N\Omega/2)}{N \sin(\Omega/2)} \right|$$

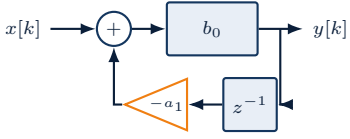
- **Sinc-like** magnitude, zeros at $f = k\, f_s/N$: cancels a known periodic disturbance exactly (choose $NT_s =$ its period).
- Linear phase, group delay $\frac{N-1}{2} T_s$.
- Poor stop-band (first side lobe only $-13\,\text{dB}$).
- Recursive form: $y[k] = y[k-1] + \frac{x[k] - x[k-N]}{N}$.

FIR versus IIR filters

FIR: only feed-forward taps



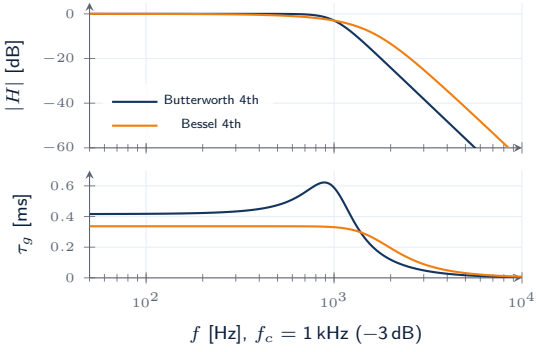
IIR: output fed back

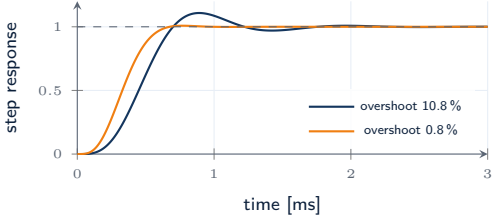


	FIR	IIR
Structure	taps only	taps + feedback
Stability	always	poles inside unit circle
Phase	linear possible (symmetric b_i)	non-linear
Order needed	high (tens–hundreds)	low (2–8)
Cost / sample	N MAC	$\sim$5 MAC per biquad
Delay	$(N-1)T_s/2$	small, f-dependent
Quantisation	robust	use biquads

In loops **IIR** wins (low delay); FIR for linear phase.

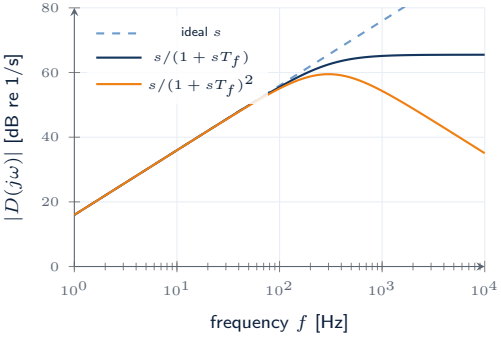
Butterworth vs. Bessel: magnitude, group delay, step response





- **Butterworth**: $|H|^2 = 1/(1+(\omega/\omega_c)^{2n})$, maximally flat magnitude; $-38\,\mathrm{dB}$ at $3f_c$; τ_g peaks near $f_c \Rightarrow$ ringing.
- **Bessel**: maximally flat group delay $\tau_g = -\mathrm{d}\varphi/\mathrm{d}\omega$, near-linear phase, clean step; only $-25\,\mathrm{dB}$ at $3f_c$.
- Butterworth for attenuation (AAF, noise), Bessel for waveform fidelity (transients).

Filtered derivative: position → velocity → acceleration



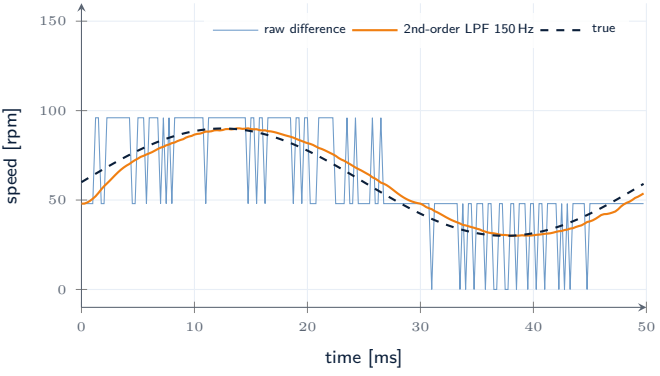
$f_f = 1/(2\pi T_f) = 300\,\text{Hz}$.

- Differentiation **amplifies** high frequencies by ω ($+20\,\text{dB/dec}$); quantisation noise is broad-band.
- Practical derivative: $D(s) = \dfrac{s}{1+sT_f}$, high-frequency gain limited to $1/T_f$.

$$\sigma_\omega = \frac{\sqrt{2}}{T_s}\,\frac{q}{\sqrt{12}}, \qquad \sigma_\alpha = \frac{\sqrt{6}}{T_s^2}\,\frac{q}{\sqrt{12}}$$

- (Backward differences; coefficient energy $1^2{+}1^2$ and $1^2{+}2^2{+}1^2$.)
- $q = 2\pi/10^4$, $T_s = 125\,\mu\text{s}$: $\sigma_\alpha \approx 2.8 \times 10^4\,\text{rad/s}^2$ — useless unfiltered.
- Better: **observer** / tracking loop (model-based, Day 3) instead of raw differences.

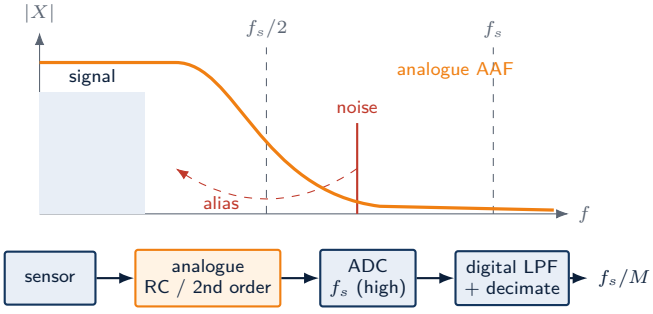
Velocity from a quantised encoder: raw vs. filtered



Simulation: $N = 2500$ lines ($\times 4$), $T_s = 125\,\mu s$, true speed 60 ± 30 rpm at 20 Hz.

- Raw $\Delta\theta / T_s$ takes only values $0, 48, 96, \ldots$ rpm.
- The LPF recovers the trend but adds **phase lag** ($\approx$1.5 ms at low f) and residual ripple.
- Loop gain multiplies this noise into the torque command $\Rightarrow$ audible noise, heating.
- Root cause is *resolution*, filtering is only a compromise.

Anti-aliasing: analogue before the ADC, digital after



- Only an **analogue** filter in front of the sampler prevents aliasing.
- Oversampling ($f_s \gg$ signal band) relaxes the analogue filter to a simple RC; the sharp filter is then done digitally before decimation.
- $\Sigma\Delta$ converters are the extreme case: very high modulator rate, sinc decimation.
- In a control loop, the analogue AAF adds delay too — keep its corner ≥ 5–$10 \times f_c$ and include it in the model.
- Synchronous sampling of PWM ripple (previous session) is a "structural" anti-aliasing measure.

Implementing filters: difference equations and biquads

General IIR (order n):

$$y[k] = \sum_{i=0}^{n} b_i\, x[k-i] - \sum_{j=1}^{n} a_j\, y[k-j]$$

Direct form II transposed biquad (per sample):

$$\begin{aligned} y &= b_0 x + s_1 \\ s_1 &= b_1 x - a_1 y + s_2 \\ s_2 &= b_2 x - a_2 y \end{aligned}$$

Higher orders: cascade of biquads (second-order sections), not one high-order polynomial.

C sketch

```
typedef struct {
    float b0,b1,b2,a1,a2,s1,s2;
} biquad_t;

static inline float biquad(biquad_t *f, float x)
{
    float y = f->b0*x + f->s1;
    f->s1   = f->b1*x - f->a1*y + f->s2;
    f->s2   = f->b2*x - f->a2*y;
    return y;
}
```

- Low f_c/f_s ($< 10^{-3}$): poles close to $z = 1 \Rightarrow$ coefficient rounding; use double / error feedback.
- Fixed point: scale to avoid overflow, watch limit cycles.
- Initialise states to steady state, reset on mode change.

Summary: signal processing and filters

- Every filter has a frequency response *and* a delay — in a loop both matter.
- Moving average: exact zeros at kf_s/N, linear phase, poor stop-band.
- FIR: stable, linear phase, costly; IIR: cheap, low delay, implement as biquads.
- Butterworth for attenuation, Bessel for waveform fidelity.
- Derivatives amplify noise $\propto \omega$; filter or, better, observe.
- Aliasing is prevented only before the ADC.

Filter selection

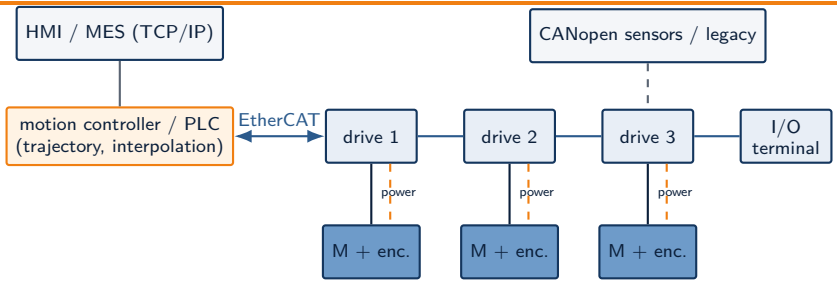
narrow resonance	notch
broad HF noise	LPF (2nd order)
offset / drift	HPF
known periodic dist.	moving avg. / notch
transient capture	Bessel
anti-aliasing	analogue LPF

[Oppenheim & Schafer, 2010]; [Ellis, 2012].

Session 6

Communication and system integration (4.5)

Communication architecture of a multi-axis machine



Level	Data	Typical cycle	Interface
Inside drive	currents, angle	31 µs to 125 µs	SPI, $\Sigma\Delta$ bit stream, BiSS / EnDat
Drive ↔ controller	set-points, actual values	0.125 ms to 4 ms	EtherCAT, PROFINET IRT, CANopen
Controller ↔ plant IT	recipes, diagnostics	10 ms to 1000 ms	Ethernet TCP/IP, OPC UA

CAN and CANopen

- **CAN** (ISO 11898): differential 2-wire bus, $120\,\Omega$ termination at both ends.
- Bitwise **arbitration**: dominant 0 wins, lower identifier = higher priority, no collisions lost.
- Classical CAN: ≤ 8 data bytes, 1 Mbit/s up to ≈40 m; CAN FD: up to 64 bytes, faster data phase.
- 15-bit CRC, ACK, error counters, bus-off.
- **CANopen** (CiA 301): object dictionary; **PDO** = real-time process data, **SDO** = configuration, NMT, SYNC, EMCY, heartbeat.

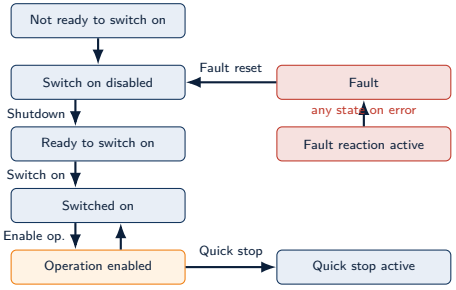


Bandwidth estimate

8-byte standard frame ≈ 111 bits (+ stuffing) $\Rightarrow$ ≈111 µs to 135 µs at 1 Mbit/s.
4 axes $\times$ (1 RxPDO + 1 TxPDO) + SYNC = 9 frames ≈ 1.1 ms.

$\Rightarrow$ CANopen suits $\geq$1 ms cycles with few axes; tight multi-axis synchronisation needs EtherCAT-class buses.

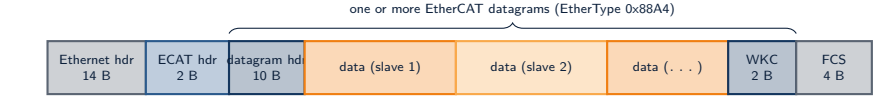
CiA 402 drive profile



Object	Meaning
0x6040	Controlword
0x6041	Statusword
0x6060 / 0x6061	Modes of operation / display
0x607A	Target position
0x6064	Position actual value
0x60FF	Target velocity
0x606C	Velocity actual value
0x6071	Target torque

- Modes: 1 PP, 3 PV, 4 PT, 6 homing, **8 CSP, 9 CSV, 10 CST**.
- Cyclic synchronous modes: the controller interpolates, the drive closes loops every bus cycle.
- Same profile over CAN, EtherCAT (CoE) and others (IEC 61800-7).

EtherCAT: processing on the fly

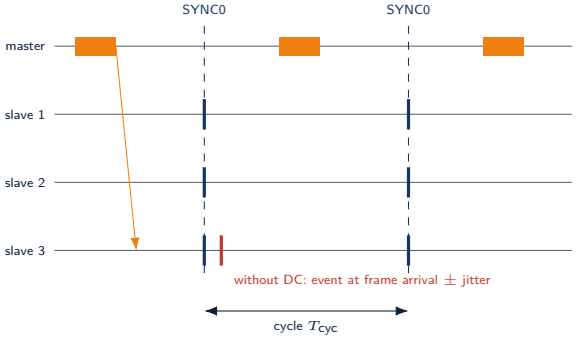


- Each slave's ESC (hardware) reads its output data and inserts its input data while the frame passes — delay well below 1 µs per node.
- **WKC** (working counter): each addressed slave increments it ⇒ the master detects missing / failed slaves.

Example: 32 axes

12 B in + 12 B out per axis, one LRW datagram: $32 \cdot 12 = 384$ B + ≈ 52 B overhead ⇒ ≈ 35 µs on the wire at 100 Mbit/s, plus node and cable delays ≈15 µs to 30 µs ⇒ fits a 100 µs cycle.

Distributed clocks, synchronisation and jitter



- **Distributed clocks (DC)**: first DC slave = reference; master measures propagation delays and compensates offset and drift of every local clock.
- All drives latch inputs and apply set-points on the same **SYNC0** event, independent of frame arrival jitter — synchronisation well below 1 µs.
- Jitter Δt appears as a position error $v\,\Delta t$: $1\,\text{m/s} \times 1\,\mu\text{s} = 1\,\mu\text{m}$.
- Latency (constant) costs phase; jitter (variable) adds noise and sub-harmonic disturbances.

Serial interfaces and fieldbus comparison

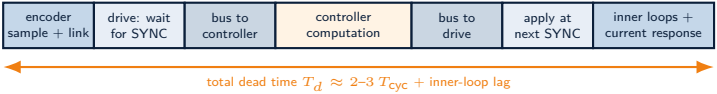
Interface	Topology	Rate	Typ. cycle / sync	Use and remarks
SPI	on-board	10 MHz to 50 MHz	per sample	ADCs, isolators; add CRC
RS-485	multi-drop	0.1–10 Mbit/s	—	Modbus RTU, encoder links
Encoder protocols	point-to-point	1 MHz to 16 MHz	every cycle	SSI, BiSS-C, EnDat; CRC
CAN / CANopen	line bus	1 Mbit/s	$\geq$1 ms	robust, cheap, slow
EtherCAT	line, on the fly	100 Mbit/s	62.5 µs–4 ms	DC sync $<$ 1 µs; CoE
PROFINET IRT	switched	100 Mbit/s	31.25 µs–ms	PROFIdrive profile



Choose the bus by **cycle time**, **synchronisation accuracy**, number of axes and drive profile, not by raw bit rate. EtherCAT slaves use a dedicated ESC chip (left). Standards: ISO 11898, IEC 61158 / 61784, IEC 61800-7, TIA/EIA-485.

EtherCAT slave controller IC: Allanchou1357, CC BY-SA 4.0, via Wikimedia Commons

Latency budget of a distributed position loop



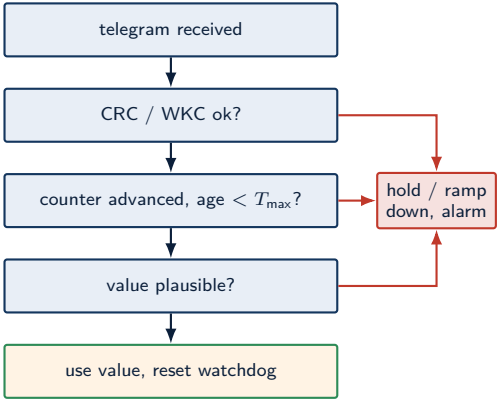
Worked example

$T_{\text{cyc}} = 250\,\mu\text{s}$, $T_d = 2.5\,T_{\text{cyc}} = 625\,\mu\text{s}$.
Position loop in the controller with $f_c = 30\,\text{Hz}$: $\Delta\varphi = 360^\circ \cdot 30 \cdot 625 \cdot 10^{-6} = 6.8^\circ$ — acceptable.
Velocity loop over the bus at $f_c = 200\,\text{Hz}$: 45° — **not acceptable**.

- Close fast loops (current, velocity) **in the drive**; send position set-points (CSP) or velocity set-points (CSV) over the bus.
- Use velocity / torque **feed-forward** values in the PDO to recover dynamics.
- Time-stamp set-points; interpolate between bus cycles inside the drive.

Data validity: CRC, watchdogs, timestamps

- **CRC**: detects transmission errors (CAN 15 bit, Ethernet FCS CRC-32, encoder CRC 6–16 bit). A CRC-protected value can still be *old*.
- **Working counter / sequence (life) counter**: detects lost, repeated or frozen telegrams.
- **Watchdogs**: SyncManager watchdog (EtherCAT), heartbeat / node guarding (CANopen), software watchdog in the drive firmware.
- **Timestamps**: age of a value $= t_{\text{now}} - t_{\text{sample}}$; reject or extrapolate stale data.
- **Plausibility**: range, rate-of-change, cross-checks (two encoders, $i_a + i_b + i_c = 0$).



Safety functions (STO, SS1, SLS) use a separate **black-channel** safety protocol (e.g. FSoE, IEC 61784-3) on top of the same bus.

Summary: communication and system integration

- CAN/CANopen: robust, deterministic arbitration, $\geq$1 ms cycles.
- CiA 402 standardises drive state machine, objects and operating modes (CSP/CSV/CST).
- EtherCAT: on-the-fly processing + distributed clocks $\Rightarrow$ 100 µs cycles, sub-µs sync.
- Latency costs phase, jitter costs accuracy: close fast loops locally.
- Validity needs CRC *and* freshness (counters, watchdogs, timestamps).

Integration checklist

- cycle time vs. loop bandwidths
- synchronisation of all sampling
- end-to-end latency budget
- error reaction for every link
- diagnostics and time-stamped logging

ETG.1000 / IEC 61158 (EtherCAT); CiA 301, CiA 402; ISO 11898.

Session 7

Day summary and course wrap-up

Day summary

3.7 Resonance

$\omega_{\mathrm{AR}} = \sqrt{k/J_L}$, $\omega_{\mathrm{R}} = \omega_{\mathrm{AR}}\sqrt{1+R}$; notch for narrow peaks, LPF for broad HF, adaptive notch for drifting modes.

4.1–4.2 Sensors and position sensing

Specify accuracy vs. resolution; $\times 4$ decoding; atan2 interpolation; RDC tracking loop; M/T velocity estimation.

4.3 Sampling

SNR $= 6.02N + 1.76$ dB; alias $= |f - kf_s|$; PWM-synchronous sampling; delay $\rightarrow 360^\circ f T_d$.

4.4 Filters

Moving average, FIR vs. IIR, Butterworth vs. Bessel, filtered derivative, analogue anti-aliasing, biquad DF-II-T.

4.5 Integration

CANopen / CiA 402, EtherCAT with DC, latency and jitter budgets, CRC + watchdog + timestamps.

One sentence

The achievable servo performance is set by mechanics (resonances) and by the measurement chain (resolution, noise, delay) — the controller can only use what they allow.

Course wrap-up: five days, one servo axis

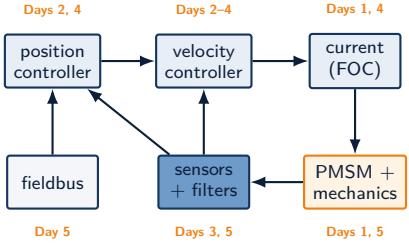
Day 1 Electrical machines & actuators; intro to modelling

Day 2 System dynamics, identification, PID, stability

Day 3 Filters, modern / adaptive / discrete control; servo intro

Day 4 Drives, FOC, cascaded loops, tuning, compensation

Day 5 Resonance, sensors, sampling, filters, fieldbus



Course wrap-up: ten principles to take home

1. Know the plant: R, L, k_t, J, B and the **mechanics** (Day 1, 5).
2. Model first, then identify and validate with FRFs (Day 2).
3. Loop shaping: $L(s) = C(s)G(s)$, ω_c, PM $\geq 45°$, GM ≥ 6 dB (Day 2).
4. Every filter and every sample is phase lag (Days 3, 5).
5. Cascade: current $\gg$ velocity $\gg$ position bandwidth, $\times$3–10 per stage (Day 4).
6. Field orientation turns the PMSM into a torque source $T = \frac{3}{2} p \psi_{\mathrm{PM}} i_q$ (Day 4).
7. Feed-forward does the work, feedback fixes the errors (Day 4).
8. Resonances are handled mechanically first, then by notch / LPF (Day 5).
9. Resolution, noise and latency of sensors bound the achievable stiffness (Day 5).
10. Synchronise and time-stamp everything; validate every datum (Day 5).

Thank you

Questions, discussion of your own axes and applications — and an invitation to measure an FRF on your next machine.

Figure credits

All plots and diagrams not listed here are original drawings computed for this course (TikZ / pgfplots; data generated by `day5/data/gen_data.py`).

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- ISO 11898-1/-2 (CAN); CiA 301 and CiA 402 (CANopen, drive profile); IEC 61800-7 (drive profiles).
- IEC 61158 / IEC 61784 and ETG.1000 (EtherCAT); IEC 60751 (industrial platinum resistance thermometers).
- Application notes on resolver-to-digital conversion, sigma-delta current sensing and encoder interfaces: Analog Devices, Texas Instruments, Infineon.